# Supplementary Material: Rare or Wrong? Constraint-Qualified Auditor Selection for Group-Preserving Visual Dataset Auditing

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## CCS CONCEPTS

• **Computing methodologies** → **Supervised learning**; **Computer vision**; • **Information systems** → **Data cleaning**.

## KEYWORDS

computer vision, visual dataset auditing, image-label noise, vision-language models, long-tailed recognition, domain shift

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## 1 OVERVIEW

This supplement makes the complete visual-auditing evidence chain auditable. It specifies how image-label pairs become frozen classifier, vision-language, self-supervised, neighborhood, and transport evidence; then defines the data split, score construction, anchor, selector, and tie-breaking rules. It reports constraint, candidate-pool, calibration-budget, operating-point, and library-size analyses, followed by activation, fallback, statistics, qualitative selection, and downstream probes. Every selector variant receives the same verified-label budget, operating point, fold construction, objective, and deterministic tie-breaking. Differences therefore isolate candidate access or qualification constraints rather than hidden supervision.

The main experiments use four datasets and five random seeds, yielding 20 dataset-seed evaluations. CIFAR-100N-LT and Food-101N-LT use natural annotation-noise evidence. DomainNet-LT and GeoDE-LT use controlled label corruption. For the controlled benchmarks, a non-corrupted example means only that the experimental corruption process did not alter its label. Only calibration error indicators may influence anchor or auditor selection; held-out error indicators enter later, after $s^\star$ is frozen.

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**Table 1: Core notation and its role in the visual-auditing pipeline.**

| Symbol | Meaning |
|---|---|
| $x_i$, $y_i$ | image and observed visual class label |
| $g_i$ | operational frequency/shift group |
| $e_i$ | verified annotation-error indicator |
| $C$ | 20% calibration subset; $e_i$ is visible to selection only here |
| $\mathcal{H}$ | 80% held-out subset; its $e_i$ is hidden until offline evaluation |
| $b_j$ | frozen base visual auditor |
| $\mathbf{z}_i$ | vector of base-auditor evidence for image $i$ |
| $\rho(\cdot)$ | normalized empirical-rank transform |
| $\mathcal{S}$ | finite singles-and-pairs candidate library |
| $s_0$ | calibration-selected baseline anchor |
| $\mathcal{F}$ | candidates satisfying every anchor-relative constraint |
| $s^\star$ | selected auditor, frozen before held-out scoring |

## 2 EXPERIMENTAL PROTOCOL AND AUDIT BOUNDARY

### 2.1 Notation and information flow

Table 1 separates visual evidence from the error indicators used for auditor selection and evaluation. After the base-evidence pipeline is fixed, the only adaptively selected output is the discrete auditor identity $s^\star$; score definitions, candidate composition, tolerances, and tie-breaking are fixed beforehand.

Operationally, the flow is *images and observed labels* → *frozen evidence* $\mathbf{z}$ → *ranked candidates* $\mathcal{S}$ → *calibration metrics on* $C$ → *feasible set* $\mathcal{F}$ → *selected auditor* $s^\star$ → *held-out review queue.* This is the audit boundary used throughout the experiments.

### 2.2 Split and evaluation semantics

For every benchmark and protocol seed in $\{0, 1, 2, 3, 4\}$, a seeded random split assigns 20% of examples to calibration before the error indicators are inspected. The remaining 80% is held out. Split membership is identical for all auditors, selector variants, candidate-pool ablations, and matched comparators within a seed. The main results therefore contain 20 paired dataset-seed evaluations.

Calibration labels determine the anchor, feasibility, and final auditor identity. For held-out reporting, each frozen auditor is separately thresholded using held-out true-error labels at a target 70% aggregate error recall. This is an offline matched-operating-point comparison: it prevents a method from appearing to reduce false positives merely by flagging fewer examples. It does not assume

that held-out error labels are available when an audit queue is deployed.

### 2.3 Benchmark construction

Table 2 lists the frozen imbalance ratio, error evidence, operational groups, and shift definition used for each benchmark.

For each seeded protocol, class counts follow an exponential long-tail schedule. Classes sorted by realized selected frequency are partitioned into three nearly equal sets named head, medium, and tail. DomainNet and GeoDE cross that frequency tier with the predefined shift status, yielding six composite groups. Region identity is retained for GeoDE evaluation but is excluded from the primary auditor scores.

DomainNet and GeoDE do not provide exhaustive natural label-error masks. In both, the protocol selects 20% of previously uncorrupted examples uniformly without replacement and replaces each label by a different class sampled from the empirical class prior. The injection location is therefore not head-only; class imbalance affects the replacement-label distribution. A non-injected example is not asserted to be independently human-verified.

### 2.4 Frozen representation and baseline settings

We use OpenCLIP `ViT-B-32` with `openai` weights (Radford et al. 8) and Hugging Face `facebook/dinov2-small` features [6]. Frozen feature batches are 128 for CLIP and 256 for DINOv2, so no encoder epochs or optimizer apply. CLIP uses eight deterministic views and prompt banks with 7/3/7/8 templates per class for CIFAR-100N/Food-101N/DomainNet/GeoDE, respectively. Frozen features use the official processors. Neighborhood support is exact cosine search on L2-normalized DINOv2 features with $k = 50$, excluding the query.

Confident-learning candidates use five-fold out-of-fold linear probes (Northcutt et al. 5) over normalized CLIP+DINOv2 features, with unweighted and class-balanced losses. Each fold uses 30 epochs, batch size 4096, AdamW, learning rate 0.03, weight decay $10^{-4}$, and a cosine schedule. AUM [7] uses a scratch ResNet-18 trained on observed labels for 100 epochs with SGD, learning rate 0.03, momentum 0.9, Nesterov updates, weight decay $5 \times 10^{-4}$, a cosine schedule, and batch size 512 for CIFAR-100N-LT or 256 otherwise. AUM preprocessing uses ImageNet normalization, CIFAR random crop with padding 4 and horizontal flip, and otherwise $224^2$ random-resized crops with scale 0.7–1.0, bicubic interpolation, and horizontal flip. CLIPCleaner [3] uses balanced logistic regression with at most 300 iterations, and SelfClean [4] follows its probability- and distance-based definitions. Table 5 lists every score entering selection.

For the linear-probe candidates, let $\phi_i$ denote the concatenated frozen feature vector and let $p_\theta(c \mid \phi_i)$ be the probe probability for class $c$. The unweighted probe minimizes cross-entropy on its training fold,

$$\mathcal{L}_{\mathrm{CE}}(\theta) = -\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \log p_\theta(y_i \mid \phi_i), \tag{1}$$

where $\mathcal{T}$ is the fold-training index set. The class-balanced probe uses the same objective with fixed class weights. Let $n_c = |\{i \in \mathcal{T} : y_i = c\}|$ be the training-fold count for class $c$, and let $\alpha_c > 0$ denote the deterministic frequency-based weight assigned to class $c$, normalized so that the mean weight over classes present in $\mathcal{T}$ is one. The weighted objective is

$$\mathcal{L}_{\mathrm{CB}}(\theta) = -\frac{1}{|\mathcal{T}|} \sum_{i \in \mathcal{T}} \alpha_{y_i} \log p_\theta(y_i \mid \phi_i). \tag{2}$$

The scratch ResNet-18 used for AUM is trained with the same observed labels; its role is only to produce the frozen AUM auditor score.

## 3 FIXED SCORE AND ANCHOR CONSTRUCTION

### 3.1 Vision-language conflict and rare-example protection

Let $q_{ic}$ be the CLIP probability for class $c$, averaged over the frozen prompt-view bank, with $\sum_c q_{ic} = 1$. The CLIP confidence and entropy scores are $-q_{i,y_i}$ and $-\sum_c q_{ic} \log q_{ic}$, respectively. Let $a_i = \arg\max_{c \neq y_i} q_{ic}$ be the most supported alternative to the observed label, and let $\gamma_i = q_{ia_i} - \max_{c \notin \{y_i, a_i\}} q_{ic}$ be the margin from the top alternative to the next alternative. With prompt-view variance $v_{ia_i}$ for the alternative class, the stable conflict score is

$$b_i = (q_{ia_i} - v_{ia_i}) + \gamma_i - q_{iy_i}. \tag{3}$$

The variance coefficient is fixed to one. Let $n_i^y$ and $n_i^a$ denote the fractions of neighbors supporting the supplied and alternative labels, and let $r_i$ be inverse-frequency rarity normalized to $[0, 1]$. The SafeClean-style score used in the baseline library is

$$s_i^{\mathrm{SC}} = b_i + 0.5 n_i^a - 0.5 n_i^y - 0.75 r_i n_i^y. \tag{4}$$

The last term protects a rare example when its local neighborhood coherently supports the supplied label. Source/region indicators are not used in the primary score.

### 3.2 Fixed non-OT, balanced-OT, and UOT expansions

For image $i$ and class $c$, the fused class cost is the mean of row-normalized visual prototype distance and CLIP image-text distance,

$$C_{ic} = \tfrac{1}{2} \widetilde{C}_{ic}^{\mathrm{vis}} + \tfrac{1}{2} \widetilde{C}_{ic}^{\mathrm{sem}}. \tag{5}$$

Write $d_i^y = C_{i,y_i}$, $a_i = \arg\min_{c \neq y_i} C_{ic}$, $d_i^a = C_{i,a_i}$, and $h_i = d_i^y - d_i^a$. Balanced and unbalanced transport follow entropic and relaxed-marginal formulations [1, 2] and use a uniform sample marginal $\mu$ and a smoothed empirical class marginal $\nu$, both nonnegative and normalized to unit mass. Balanced OT solves

$$P^{\mathrm{OT}} = \arg\min_{P \geq 0} \left[ \langle P, C \rangle + \varepsilon \sum_{i,c} P_{ic} (\log P_{ic} - 1) \right], \tag{6}$$
$$\text{s.t.} \quad P\mathbf{1} = \mu, \quad P^\top \mathbf{1} = \nu.$$

Fixed UOT instead relaxes the marginal constraints:

$$P^{\mathrm{UOT}} = \arg\min_{P \geq 0} \Big[ \langle P, C \rangle + \varepsilon \sum_{i,c} P_{ic} (\log P_{ic} - 1) + \tau_x \, \mathrm{KL}(P\mathbf{1} \,\|\, \mu) + \tau_y \, \mathrm{KL}(P^\top \mathbf{1} \,\|\, \nu) \Big]. \tag{7}$$

**Table 2: Frozen benchmark protocols. IR is the requested head-to-tail imbalance ratio. Natural means the released annotation/verification signal is used; controlled means that a known synthetic corruption mask is generated by label replacement and then partitioned under the same calibration/held-out protocol.**

| Benchmark | IR | Error evidence | Operational groups | Shift definition |
|---|---|---|---|---|
| CIFAR-100N-LT | 50 | Natural human noise | head/medium/tail | none |
| Food-101N-LT | 20 | Natural verification | head/medium/tail | none |
| DomainNet-LT | 50 | 20% controlled | frequency tier × source/shift | real is source; clipart, infograph, painting, quickdraw, and sketch are shifted |
| GeoDE-LT | 20 | 20% controlled | frequency tier × source/shift | Americas and Europe are reference; Africa, East Asia, Southeast Asia, and West Asia are shifted |

Here $\mathrm{KL}(a \,\|\, b) = \sum_j a_j \log(a_j/b_j) - a_j + b_j$ is the generalized KL divergence for nonnegative vectors. After row-normalizing the selected transport plan $P$, the mass gap is

$$m_i = \frac{P_{i,a_i}}{\sum_c P_{ic}} - \frac{P_{i,y_i}}{\sum_c P_{ic}}. \tag{8}$$

With robust median/MAD standardization $z(\cdot)$, every fixed expansion uses

$$s_i^{\mathrm{Bi}} = 0.5z(b_i) + z(h_i) + z(m_i) + 0.35n_i^a - 0.35n_i^y - 0.75r_i n_i^y. \tag{9}$$

The no-transport variant replaces plan masses with fixed sigmoid transforms of $d_i^y$ and $d_i^a$. Balanced OT enforces both marginals; fixed UOT uses entropy $\varepsilon = 0.05$, marginal relaxation $\tau_x = \tau_y = 0.25$, at most 300 Sinkhorn iterations, and tolerance $10^{-7}$. These numerical settings and all coefficients are frozen; no transport variant is tuned inside the final 210-candidate selector.

### 3.3 Exact anchor definition

The anchor is selected only from the 17 baseline scores and their pairwise rank means. On calibration data, let $\mathcal{G}_T$ contain tail groups, $\mathcal{G}_S$ contain shifted groups, and $\mathrm{gap}(s) = \max_g F_g(s) - \min_g F_g(s)$. The frozen anchor objective is

$$J_0(s) = 1 - A(s) + \max_{g\in\mathcal{G}_T} F_g(s) + \max_{g\in\mathcal{G}_S} F_g(s) + 0.25\,\mathrm{gap}(s), \tag{10}$$

where a term is omitted when the corresponding group type is absent or undefined. The minimizing baseline candidate becomes $s_0$; deterministic library order resolves an exact tie. The three BiSafe expansion scores cannot become the anchor. The matched adaptive baseline then starts from this same $s_0$ and runs the complete qualification rule with access restricted to baseline candidates.

## 4 RARE-OR-WRONG MOTIVATING EVIDENCE

The main paper's RQ1 reports false-positive amplification on difficult valid examples at the matched 70% error-recall operating point: values above one mean that valid tail or shifted examples are flagged more often than the pooled valid population. Figure 1 gives the complete per-auditor, per-dataset breakdown underlying those numbers.

## 5 EXACT ADAPTIVE SELECTOR

### 5.1 Exact Selector Specification

*Data flow.* The implemented pipeline is *image-label records → frozen classifier, vision-language, self-supervised, neighborhood, and transport evidence → rank normalization and pairwise composition → verified-calibration metrics → pooled feasibility gate → fold-deletion stability gate → worst-group-FPR minimization → deterministic fallback*. The resulting auditor is then frozen before any held-out metric is computed. This ordering is operational, not merely expository: held-out error indicators cannot affect candidate construction, feasibility, or tie-breaking.

Let $\mathbf{z}_i = [b_1(i), \ldots, b_B(i)]^\top$ collect the frozen base-auditor scores for image $i$, let $\mathcal{S}$ denote the finite candidate set, and let $s_0$ denote the calibrated anchor. Individual auditor scores are normalized by empirical rank. For an index set $\mathcal{I}$ and score vector $u$, the normalized rank is

$$\rho_{\mathcal{I}}[u_i] = \frac{1}{|\mathcal{I}| - 1}\left(\sum_{\ell\in\mathcal{I}} \mathbf{1}\{u_\ell < u_i\} + \frac{1}{2}\sum_{\ell\in\mathcal{I}} \mathbf{1}\{u_\ell = u_i\} - \frac{1}{2}\right), \tag{11}$$

with rank 0.5 assigned when $|\mathcal{I}| = 1$. Larger suspicion scores therefore receive larger ranks, and ties receive their average rank. For two auditors $s_a$ and $s_b$, a pairwise candidate is

$$s_{a,b}(i) = \frac{1}{2}\left[\rho_{\mathcal{I}}[s_a(i)] + \rho_{\mathcal{I}}[s_b(i)]\right]. \tag{12}$$

The final expanded pool contains the baseline candidate pool together with the fixed non-OT, balanced-OT, and fixed-UOT candidates. Separately tuned or calibration-optimized transport variants are excluded from the final candidate set to avoid nested selection bias.

The final selector uses

$$\epsilon_A = 0.005, \qquad \epsilon_R = 0.02, \qquad K = 5. \tag{13}$$

For every candidate $s$, we compute pooled calibration AUPRC $A(s)$, worst-group error recall $R_{\mathrm{wg}}(s)$, and the worst-group false-positive rate $F_{\mathrm{wg}}(s)$. We additionally construct five leave-one-fold-out calibration subsets and compute $R_{\mathrm{wg}}^{(-k)}(s)$.

Candidate $s$ is feasible only if

$$A(s) \geq A(s_0) - 0.005, \tag{14}$$

$$R_{\mathrm{wg}}(s) \geq R_{\mathrm{wg}}(s_0) - 0.02, \tag{15}$$

$$R_{\mathrm{wg}}^{(-k)}(s) \geq R_{\mathrm{wg}}^{(-k)}(s_0) - 0.02, \qquad \forall k. \tag{16}$$

Selection among feasible candidates is lexicographic:

(1) minimize calibration worst-group FPR;
(2) among ties, maximize calibration AUPRC;
(3) among remaining ties, prefer the anchor;

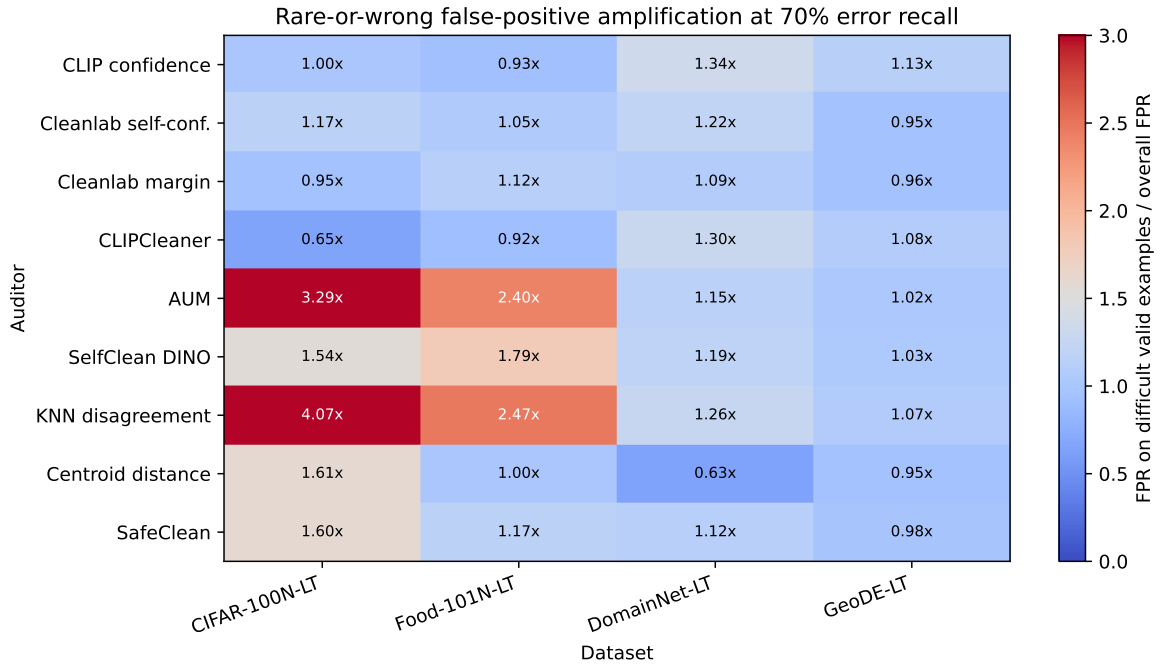



**Figure 1: Rare-or-wrong false-positive amplification. Each cell reports difficult valid-example FPR divided by overall FPR at the matched 70% error-recall operating point. Tail classes are the difficult groups for CIFAR-100N-LT and Food-101N-LT; shifted groups are difficult for DomainNet-LT and GeoDE-LT.**

(4) use deterministic candidate ordering as the final tie-breaker.

Because the anchor belongs to $\mathcal{S}$, it always satisfies the anchor-relative inequalities for non-negative tolerances. Therefore the calibration feasible set is non-empty.

*Pseudocode.* Given verified calibration set $C$, anchor $s_0$, and candidate pool $\mathcal{S}$:

(1) compute pooled metrics for all $s \in \mathcal{S}$;
(2) construct five leave-one-fold-out calibration subsets;
(3) remove candidates violating either pooled constraint;
(4) remove candidates violating worst-group recall on any leave-one-fold-out subset;
(5) minimize worst-group FPR over the remaining candidates;
(6) apply deterministic tie-breaking;
(7) return the selected auditor.

No held-out evaluation label is used during selection.

## 6 SELECTOR CONSTRAINT ABLATION

The main paper compares the calibrated anchor, ranking-only selection, unconstrained worst-group-FPR selection, pooled anchor-relative constraints, and the complete leave-one-fold-out selector.

Across 20 evaluations, ranking-only and unconstrained WG-FPR selection each produce seven held-out downside events. Pooled constraints reduce this to four and the complete selector to two.

The downside criterion was fixed before the final held-out comparison:

$$\Delta\text{AUPRC} < -0.005 \quad \text{or} \quad \Delta R_{\text{wg}} < -0.02. \tag{17}$$

These thresholds were not selected after observing the final results.

Table 3 reports the full paired prevented/introduced downside counts, exact and Holm-corrected $p$-values, and hierarchical bootstrap intervals underlying this comparison.

## 7 STATISTICAL INTERPRETATION

The paired exact comparisons are underpowered after Holm correction, so the statistics should be read as effect-size evidence rather than a proof of uniform superiority. They support two narrower observations. First, unconstrained selection can produce substantial group-level downside despite favorable aggregate ranking behavior. Second, the complete constraint set behaves more conservatively in the evaluated regimes.

For example, defining FPR improvement as comparator WG-FPR minus BiSafeAudit WG-FPR, the mean improvement over ranking-only selection is 0.009797, with hierarchical bootstrap interval $[0.002086, 0.019070]$; mean AUPRC is 0.004499 lower. This is the intended trade-off rather than an ambiguity in sign: the method sacrifices some pooled ranking performance to reduce observed worst-group false positives.

## 8 ACTIVATION AND EXACT FALLBACK

Table 4 and Figure 2 show the per-dataset activation and fallback counts. The selector activates the expanded candidate pool in seven of twenty runs and retains the anchor in the other thirteen. DomainNet-LT is the clearest fallback regime: the complete selector

**Table 3: Paired statistical analysis of the full selector relative to alternative selection regimes across 20 dataset-seed evaluations. "Prevented" counts cases with a downside under the comparator but not under BiSafeAudit; "introduced" denotes the reverse. Exact paired tests are Holm-corrected. Bootstrap intervals are hierarchical 95% confidence intervals.**

| Comparator vs. BiSafeAudit | Prevented | Introduced | Exact $p$ | Holm $p$ |
|---|---|---|---|---|
| Ranking-only | 6 | 1 | 0.1250 | 0.2500 |
| Unconstrained WG-FPR | 5 | 0 | 0.0625 | 0.1875 |
| Pooled-only | 2 | 0 | 0.5000 | 0.5000 |

| Comparator vs. BiSafeAudit | ΔAUPRC | ΔWG Recall | WG-FPR improvement |
|---|---|---|---|
| Ranking-only | −0.004499 [−0.011174, −0.000636] | +0.006292 [−0.016040, +0.034260] | +0.009797 [+0.002086, +0.019070] |
| Unconstrained WG-FPR | +0.002400 [−0.001957, +0.008252] | +0.002391 [−0.013980, +0.020713] | −0.001443 [−0.007573, +0.002210] |
| Pooled-only | 0.000000 [−0.002087, +0.002706] | −0.001214 [−0.011839, +0.010990] | −0.002184 [−0.007452, 0.000000] |

**Table 4: Constraint-qualified activation of the expanded candidate pool. Activation denotes selection of a non-anchor candidate; fallback denotes retention of the calibrated anchor.**

| Dataset | Activation | Fallback |
|---|---|---|
| CIFAR-100N-LT | 4/5 | 1/5 |
| Food-101N-LT | 1/5 | 4/5 |
| DomainNet-LT | 0/5 | 5/5 |
| GeoDE-LT | 2/5 | 3/5 |
| Overall | 7/20 | 13/20 |

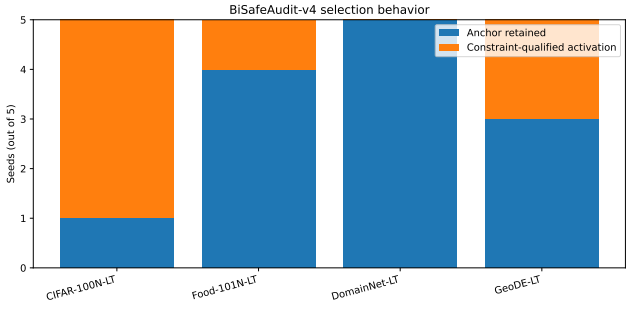



**Figure 2: Activation and fallback behavior across datasets. "Activation" denotes selection of a non-anchor auditor satisfying all final constraints.**

retains the anchor in all five seeds. Activation is optional by design; exact fallback is part of the method, not a failure mode.

Figure 3 gives the complementary activation-risk view. Conditional downside is lowest for the complete selector: 28.6% (2/7), versus 36.8% for ranking-only selection (7/19), 53.8% for unconstrained WG-FPR selection (7/13), and 44.4% for pooled-only selection (4/9).

## 9 FEASIBILITY CASE STUDIES

The main paper summarizes the selection behavior quantitatively. Here we add a diagnostic feasibility case study, shown in Figure 4, covering three representative outcomes: conservative anchor fallback, a candidate that satisfies the pooled constraints but is rejected by the leave-one-fold-out stability condition, and constraint-qualified activation.

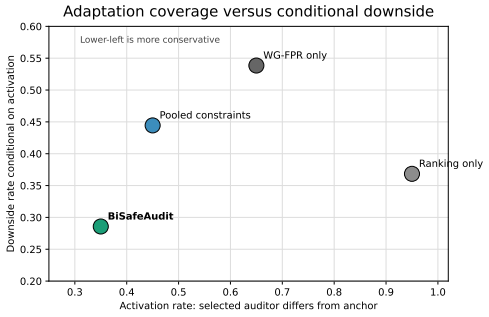



**Figure 3: Activation coverage versus downside rate conditional on activation.**

The examples illustrate why pooled calibration metrics alone are insufficient. A candidate can appear feasible under pooled evaluation but lose worst-group recall when a calibration fold is removed.

## 10 EXACT CANDIDATE-POOL ABLATION

The candidate-pool experiment reconstructs selection under four exact pools: baseline-only, baseline plus non-OT expansion, baseline plus balanced/fixed OT candidates, and the complete pool.

On CIFAR-100N-LT, the non-OT expansion attains the highest mean AUPRC, whereas OT-accessible pools make a different trade-off between ranking quality and group preservation. On GeoDE-LT, the adaptive improvement observed for the complete selector depends on access to the OT family. Food-101N-LT and DomainNet-LT remain essentially at their baseline operating points.

These results motivate treating OT as complementary evidence rather than as a universally dominant auditor. Table 5 lists the full verified candidate library underlying every pool.

Figure 5 gives the full comparison grid across candidate pools and datasets, and Figure 6 summarizes the same ablation as a heatmap.

## 11 CALIBRATION-LABEL BUDGET

Table 6 reports the complete per-dataset, per-budget breakdown. The smallest verification budgets can be unstable: 1% to 2% calibration produces noticeably larger variability in several settings. Behavior becomes substantially more stable once the calibration set reaches approximately 5% to 10% in the evaluated benchmarks.

Increasing the verified fraction to 20% does not provide a consistent monotonic improvement. We therefore regard 5% to 10% as

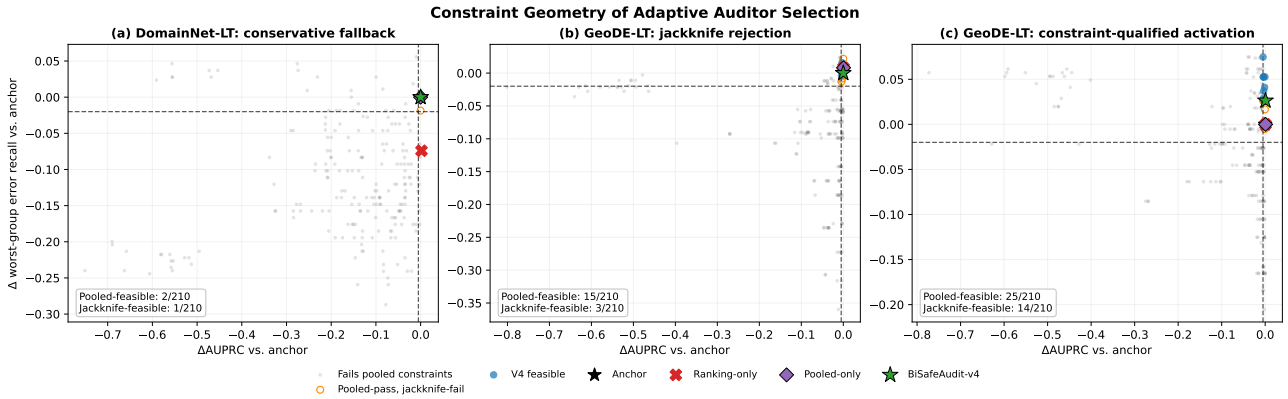



**Figure 4: Feasibility case study for the final selector. The panels separate fallback, stability rejection, and qualified activation, making clear that pooled calibration performance alone is not the implemented decision rule.**

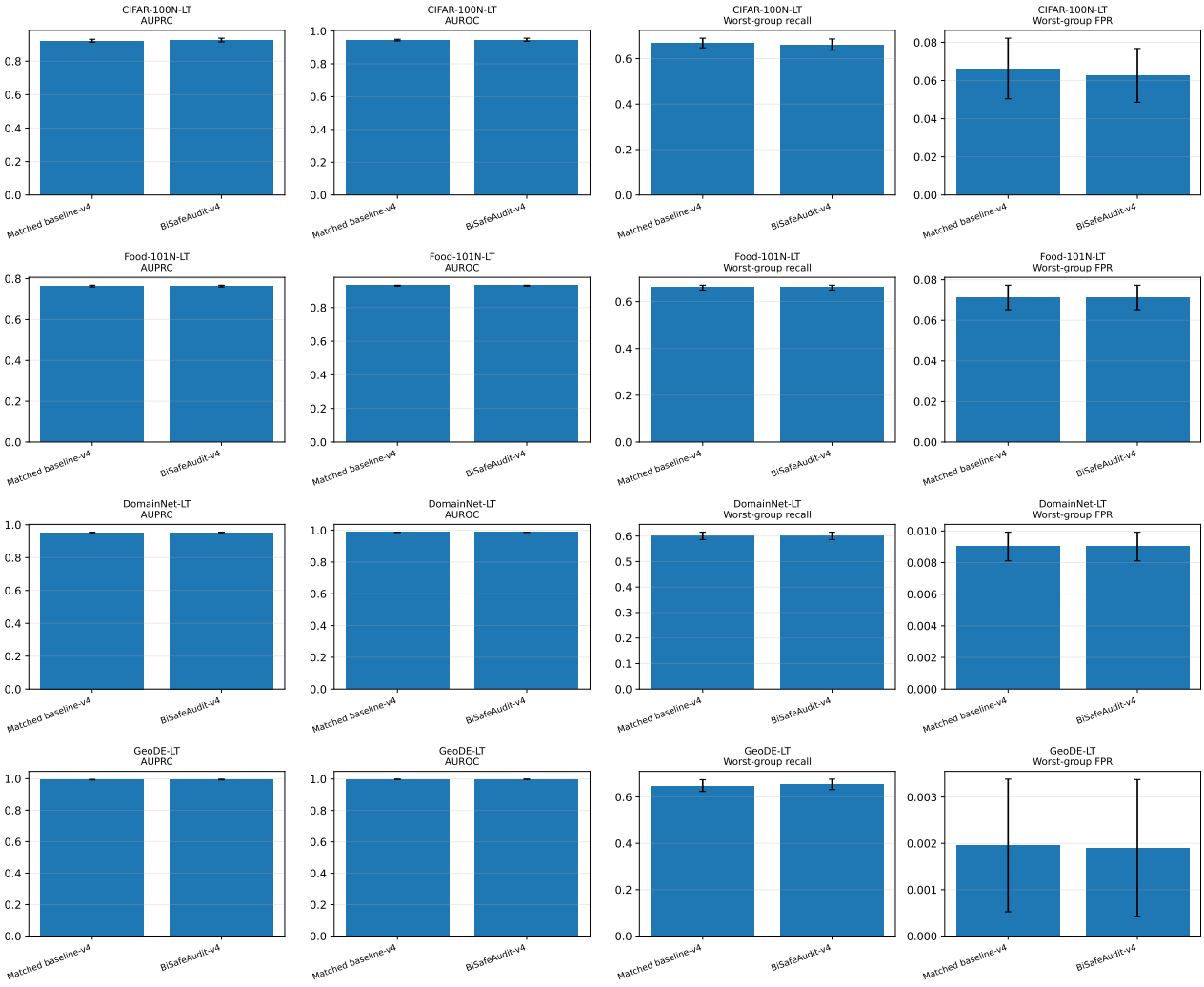



**Figure 5: Full comparison grid used to inspect ranking and group-preservation behavior across candidate pools and datasets.**

**Table 5: Verified candidate library for the selector. Every base score is unit-rank normalized before selection. Within each pool, the implementation includes each single score and every equal-weight pairwise rank mean. Thus the full pool contains 20 single scores and 190 pairwise rank means, for 210 candidates.**

| Base score | Family | Matched | +Non-OT | +OT | Full | OT | Score definition |
|---|---|---|---|---|---|---|---|
| clip_inverse_confidence | Vision-language confidence | yes | yes | yes | yes | no | Negative given-label support from the frozen prompt/view CLIP probability summary. |
| clip_entropy | Vision-language uncertainty | yes | yes | yes | yes | no | Predictive entropy of the frozen prompt/view CLIP probability summary. |
| stable_alt_conflict | Vision-language label conflict | yes | yes | yes | yes | no | Stable alternative-class evidence plus alternative gap minus given-label support. |
| safeclean | SafeClean-style hybrid evidence | yes | yes | yes | yes | no | SafeClean score from prompt/view summary, neighbor agreement, and rarity protection without source or region identity. |
| cleanlab_fused_none_self_confidence | Confident learning | yes | yes | yes | yes | no | Out-of-fold self-confidence error score from fused probabilities. |
| cleanlab_fused_none_normalized_margin | Confident learning | yes | yes | yes | yes | no | Out-of-fold normalized-margin error score from fused probabilities. |
| cleanlab_fused_balanced_self_confidence | Confident learning | yes | yes | yes | yes | no | Class-balanced out-of-fold self-confidence error score from fused probabilities. |
| cleanlab_fused_balanced_normalized_margin | Confident learning | yes | yes | yes | yes | no | Class-balanced out-of-fold normalized-margin error score from fused probabilities. |
| clipcleaner_zero_gmm | CLIPCleaner | yes | yes | yes | yes | no | GMM suspiciousness from zero-shot CLIP probabilities. |
| clipcleaner_zero_consistency | CLIPCleaner | yes | yes | yes | yes | no | Label-consistency suspiciousness from zero-shot CLIP probabilities. |
| clipcleaner_visual_gmm | CLIPCleaner | yes | yes | yes | yes | no | GMM suspiciousness from visual CLIPCleaner probabilities. |
| clipcleaner_visual_consistency | CLIPCleaner | yes | yes | yes | yes | no | Label-consistency suspiciousness from visual CLIPCleaner probabilities. |
| clipcleaner_combined | CLIPCleaner | yes | yes | yes | yes | no | Combined zero-shot and visual CLIPCleaner suspiciousness. |
| aum_resnet18_scratch | Training dynamics | yes | yes | yes | yes | no | Area-under-the-margin error score from a scratch ResNet-18 training run. |
| selfclean_dinov2 | Self-supervised geometry | yes | yes | yes | yes | no | SelfClean-style score from DINOv2 feature distances. |
| knn_disagreement | Neighborhood agreement | yes | yes | yes | yes | no | One minus the local neighbor support for the given label. |
| centroid_distance | Representation geometry | yes | yes | yes | yes | no | Distance from each sample to the prototype of its given label. |
| bisafeot_no_ot | BiSafeOT ablation | no | yes | no | yes | no | BiSafe evidence score with transport disabled. |
| bisafeot_balanced_ot | BiSafeOT transport | no | no | yes | yes | balanced OT | BiSafe evidence score with balanced optimal transport. |
| bisafeot_fixed_uot | BiSafeOT transport | no | no | yes | yes | unbalanced OT | BiSafe evidence score with the fixed unbalanced OT configuration. |

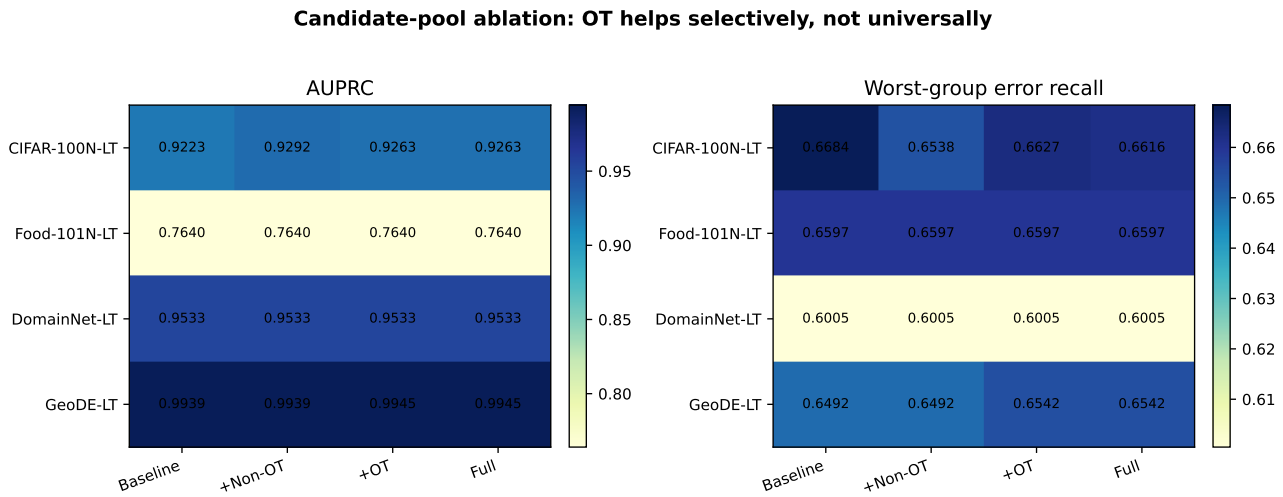



**Figure 6: Candidate-pool ablation summary. The heatmap shows that OT-accessible candidates help selectively, especially for GeoDE-LT group recall, but are not uniformly dominant across datasets or metrics.**

a practical range observed in these experiments, rather than as a theoretically optimal budget.

## 12 POST-HOC ROBUSTNESS AND SENSITIVITY ANALYSES

All analyses in this section were conducted after freezing the headline selector configuration: $\epsilon_A = 0.005$, $\epsilon_R = 0.02$, and target error recall 0.70. They are therefore robustness checks rather than tuning experiments. Held-out downside keeps the main-paper definition: $\Delta\text{AUPRC} < -0.005$ or $\Delta R_{\text{wg}} < -0.02$ relative to the calibrated anchor.

**Table 6: Complete calibration-label budget study over five seeds. Δ quantities are measured relative to the matched selector under the same calibration budget. Positive FPR improvement denotes lower worst-group false-positive rate.**

| Dataset | Budget | AUPRC | WG Recall | WG-FPR | Act. | AUPRC down. | WG-R down. | ΔAUPRC / ΔWG-R / FPR imp. |
|---|---|---|---|---|---|---|---|---|
| CIFAR-100N-LT | 1% | .9233±.0159 | .6390±.0463 | .0751±.0189 | 1/5 | 0/5 | 0/5 | +.000203/+.012766/+.005844 |
| | 2% | .9189±.0271 | .6633±.0201 | .0800±.0394 | 2/5 | 0/5 | 0/5 | +.006980/+.003404/+.012841 |
| | 5% | .9267±.0166 | .6542±.0184 | .0733±.0305 | 3/5 | 0/5 | 1/5 | +.000485/+.006092/+.020145 |
| | 10% | .9318±.0076 | .6677±.0166 | .0517±.0077 | 4/5 | 0/5 | 0/5 | +.010471/+.013096/+.017521 |
| | 20% | .9263±.0112 | .6616±.0245 | .0627±.0141 | 4/5 | 0/5 | 1/5 | +.004009/−.006757/+.003618 |
| Food-101N-LT | 1% | .7241±.0339 | .6655±.0278 | .0898±.0157 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 2% | .7396±.0193 | .6816±.0128 | .0865±.0205 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 5% | .7630±.0028 | .6709±.0181 | .0723±.0057 | 2/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 10% | .7637±.0042 | .6650±.0182 | .0705±.0070 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 20% | .7640±.0043 | .6597±.0105 | .0713±.0060 | 1/5 | 0/5 | 1/5 | .000000/.000000/.000000 |
| DomainNet-LT | 1% | .9524±.0020 | .5759±.0296 | .0151±.0107 | 1/5 | 0/5 | 0/5 | −.000805/−.001818/+.000940 |
| | 2% | .9527±.0011 | .5860±.0323 | .0112±.0038 | 1/5 | 0/5 | 0/5 | +.004472/−.002452/+.002148 |
| | 5% | .9531±.0010 | .5994±.0168 | .0107±.0038 | 1/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 10% | .9533±.0012 | .6005±.0146 | .0090±.0009 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 20% | .9533±.0012 | .6005±.0146 | .0090±.0009 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| GeoDE-LT | 1% | .9869±.0134 | .6299±.0584 | .0053±.0049 | 0/5 | 0/5 | 0/5 | .000000/.000000/.000000 |
| | 2% | .9933±.0019 | .6548±.0327 | .0028±.0023 | 1/5 | 0/5 | 0/5 | +.000316/+.003207/.000000 |
| | 5% | .9943±.0032 | .6539±.0386 | .0017±.0018 | 2/5 | 0/5 | 0/5 | +.000982/+.001658/+.000059 |
| | 10% | .9948±.0004 | .6517±.0300 | .0020±.0014 | 2/5 | 0/5 | 0/5 | +.001580/+.007461/+.000070 |
| | 20% | .9945±.0024 | .6542±.0224 | .0019±.0015 | 2/5 | 0/5 | 0/5 | +.000625/+.005053/+.000059 |

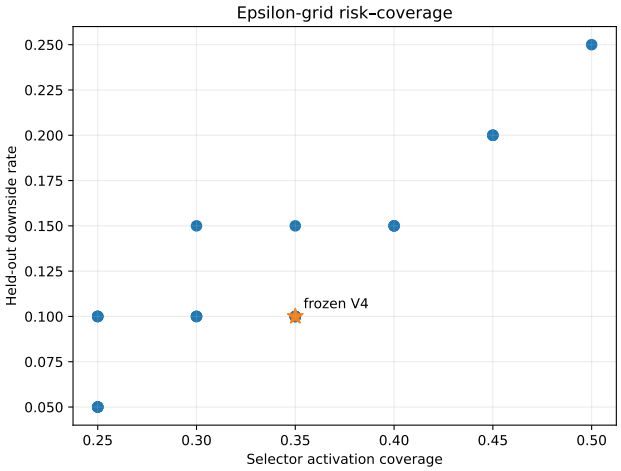



**Figure 7: Post-hoc feasibility-tolerance risk-coverage grid. Looser tolerances generally increase constraint-qualified activation, but also expose more held-out downside. The frozen setting $(\epsilon_A, \epsilon_R) = (0.005, 0.02)$ activates in 7 of 20 runs and incurs 2 of 20 downside events.**

## 12.1 Feasibility-Tolerance Sensitivity

Figure 7 sweeps the feasibility tolerances $(\epsilon_A, \epsilon_R)$ and Table 7 reports a one-dimensional slice through this grid at the frozen $\epsilon_R = 0.02$.

## 12.2 Target-Recall Operating-Point Sensitivity

Table 8 repeats the full selector at 60%, 70%, and 80% target error recall with the tolerances held fixed.

**Table 7: Tolerance slice at $\epsilon_R = 0.02$. The nearby $\epsilon_A = 0.0025$ and $\epsilon_A = 0.005$ settings have identical aggregate behavior, supporting local stability of the frozen point rather than a single isolated threshold.**

| $\epsilon_A$ | Activation | Downside | AUPRC down. | WG-recall down. |
|---|---|---|---|---|
| 0 | 6/20 | 2/20 | 0/20 | 2/20 |
| 0.0025 | 7/20 | 2/20 | 0/20 | 2/20 |
| 0.005 | 7/20 | 2/20 | 0/20 | 2/20 |
| 0.01 | 7/20 | 2/20 | 1/20 | 1/20 |
| 0.02 | 9/20 | 4/20 | 3/20 | 1/20 |

**Table 8: Operating-point sensitivity with the frozen tolerances. The qualitative conclusion persists across target error-recall settings of 60%, 70%, and 80%, but we do not claim complete operating-point invariance.**

| Target recall | Activation | Downside | AUPRC dn. | WG-R dn. | FPR imp. |
|---|---|---|---|---|---|
| 60% | 8/20 | 2/20 | 0/20 | 2/20 | +0.002281 |
| 70% | 7/20 | 2/20 | 0/20 | 2/20 | +0.001386 |
| 80% | 5/20 | 2/20 | 0/20 | 2/20 | +0.003360 |

## 12.3 Group-Metric Definedness

Table 9 counts undefined FPR, recall, and precision values across all 90 dataset-group metric instances at the frozen 70% operating point.

## 12.4 Candidate-Library-Size Sensitivity

Table 10 reports activation and downside rates as the candidate library is grown from 5 to 20 base scores under deterministic nested subset orderings.

# 13 SAFETY-CURVE VISUALIZATIONS

Figure 8 shows diagnostic ranking and group-preservation curves across auditor choices and review operating points.

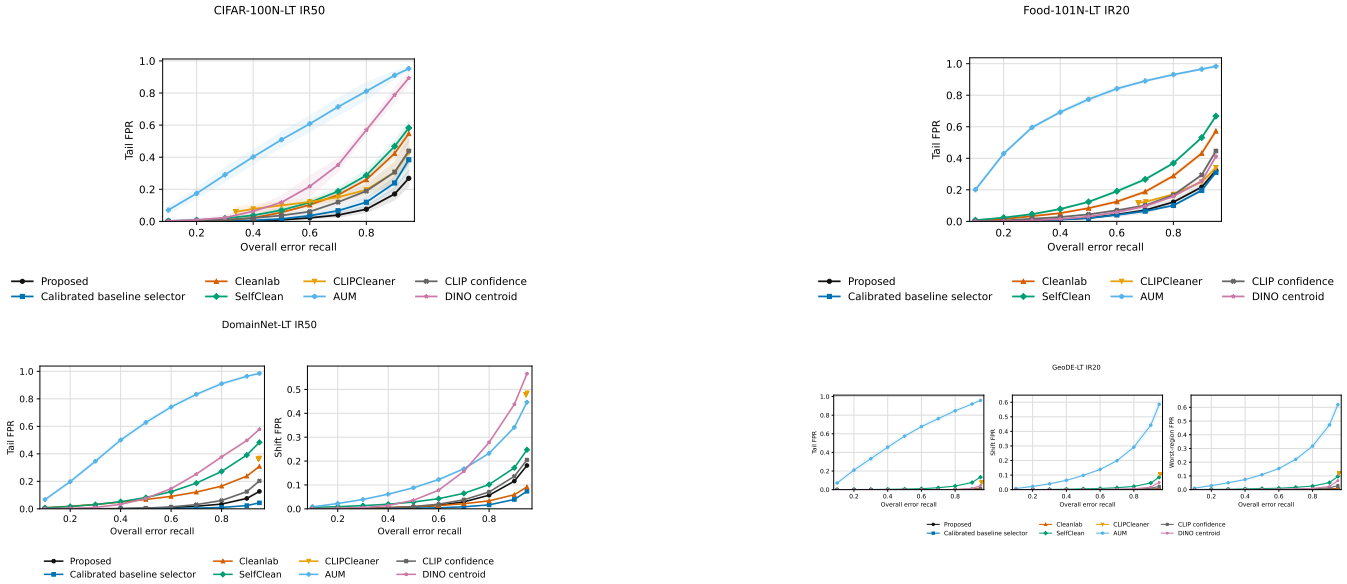



**Figure 8: Dataset-specific safety curves for aggregate ranking and group-preserving review behavior.**

**Table 9: Undefined group-metric frequency at the frozen 70% operating point. Worst-group evaluation did not rely on silently excluding undefined groups in the reported evaluation.**

| Dataset | Groups | Und. FPR | Und. recall | Und. prec. |
|---|---|---|---|---|
| CIFAR-100N-LT | 15 | 0 | 0 | 0 |
| Food-101N-LT | 15 | 0 | 0 | 0 |
| DomainNet-LT | 30 | 0 | 0 | 0 |
| GeoDE-LT | 30 | 0 | 0 | 0 |
| Total | 90 | 0 | 0 | 0 |

**Table 10: Post-hoc candidate-library-size sensitivity. Frozen anchor components are retained in every reduced pool, and remaining auditors are added using five deterministic nested subset orderings. Smaller pools activate less often, so their lower downside should not be interpreted as intrinsic safety.**

| Scores/cand. | Activation | Downside | Full agr. | FPR imp. |
|---|---|---|---|---|
| 5/15 | 0.080 [0.000,0.150] | 0.030 [0.000,0.050] | 0.660 | +0.000607 |
| 10/55 | 0.200 [0.150,0.250] | 0.080 [0.050,0.100] | 0.730 | +0.001766 |
| 15/120 | 0.300 [0.250,0.350] | 0.100 [0.100,0.100] | 0.850 | +0.001470 |
| 20/210 | 0.350 [0.350,0.350] | 0.100 [0.100,0.100] | 1.000 | +0.001386 |

## 14 QUALITATIVE SELECTION PROTOCOL

Qualitative cases were not chosen by manually browsing the dataset first. Candidate ordering was fixed using held-out auditor-score criteria before image inspection.

For preservation examples, the predetermined criterion required a non-corrupted rare or shifted example for which unconstrained ranking assigned high suspicion while the final selector assigned a lower suspicion rank. Candidates were then inspected sequentially in this fixed order.

Visual inspection was used only for two purposes:

(1) verifying that the displayed image was semantically compatible with its canonical benchmark label; and
(2) excluding unusable images, such as samples dominated by severe watermarks or other overlays.

The first candidate passing these checks was retained. The scoring criterion was not relaxed after inspection.

This procedure also exposed an important limitation of benchmarks using controlled corruption. A label can be marked as "not corrupted" by the experimental protocol while still appearing semantically questionable to a human observer. We therefore avoid describing every such sample as independently verified clean data.

The main paper contains the primary qualitative panel. To avoid duplicating that visual evidence, Figure 9 reports a separate set of additional candidate examples from later ranks in the same fixed candidate ordering. These examples are not used as new quantitative evidence; they illustrate the visual diversity of high-ranked candidate cases after the ranking and filtering protocol had already been fixed.

## 15 DOWNSTREAM PROBE ANALYSIS

For downstream utility, frozen DINOv2 linear probes use 50 epochs, batch size 4096, AdamW, learning rate 0.01, weight decay $10^{-4}$, and a cosine schedule.

On CIFAR-100N-LT, probe accuracy rises from 0.7160 without cleaning to 0.7218 after BiSafeAudit cleaning, while the matched adaptive baseline reaches 0.7261. Thus, BiSafeAudit improves over no cleaning but does not outperform the matched baseline in this setting.

On DomainNet-LT, accuracy increases from 0.5928 without cleaning to 0.6039 for both the matched selector and BiSafeAudit.

For GeoDE-LT, overall accuracy changes from 0.8969 without cleaning to 0.8951 with BiSafeAudit. Tail accuracy, however, rises

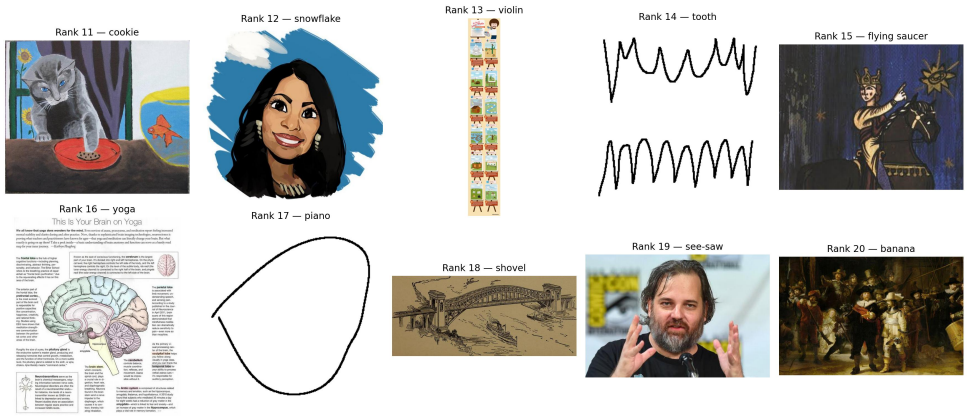



**Figure 9: Additional qualitative candidate examples from the fixed qualitative candidate list, inspected only after all scoring and selection rules were frozen.**

from 0.8656 to 0.8920, consistent with the group-preservation motivation.

Because these results are mixed and use frozen-feature probes, downstream accuracy is treated as secondary evidence rather than the primary claim of the paper.

## 16 REPRODUCIBILITY NOTES

The final selector and evaluation use:

- protocol seeds 0-4 for every benchmark;
- one seeded 20% calibration/80% held-out split per benchmark-seed, shared by every compared method;
- a 70% matched error-recall operating point;
- five leave-one-fold-out calibration splits;
- AUPRC tolerance $\epsilon_A = 0.005$;
- worst-group error-recall tolerance $\epsilon_R = 0.02$;
- worst-group false-positive rate as the feasible-set objective;
- anchor preference in tied feasible solutions;
- deterministic candidate ordering as the final tie-breaker.

The matched baseline uses the same calibration budget, folds, tolerances, objective, and tie-breaking rules. Only candidate-pool access differs. Each run stores deterministic candidate names and ordering, calibration membership, and the selected candidate. No held-out evaluation label is used to choose the anchor or final auditor, tune a score coefficient, relax a constraint, or resolve a tie.

*Selector complexity.* Let $N$ denote the number of calibration examples, $M$ the number of candidate auditors, and $K$ the number of stability folds. Re-ranking all candidates on the pooled calibration set and the $K$ leave-one-fold-out subsets has a worst-case time complexity of $O((K+1)MN \log N)$ and requires $O(MN)$ score storage when candidate vectors are cached. After ranking, foldwise metric evaluation and qualification require $O((K+1)MN)$ operations. In the reported protocol, $M = 210$ and $K = 5$. These bounds describe the finite selector stage and exclude the one-time upstream feature extraction and auxiliary-model computation. Deployment-scale wall-clock benchmarking is left for future work.

## 17 ADDITIONAL LIMITATIONS

The four benchmarks do not cover every real-world data-quality failure. Controlled corruption permits exact recovery measurement but does not reproduce every property of naturally occurring annotation noise. The finite 210-candidate library may overfit a limited verified set; the library-size analysis measures empirical sensitivity but does not supply a complexity bound.

The leave-one-fold-out gate checks only worst-group error recall. It is a local calibration stability heuristic, not a simultaneous confidence bound for recall, AUPRC, or FPR and not a population guarantee. Moreover, matched-recall held-out FPR is an offline comparison requiring evaluation error labels; an operational system would need a separately calibrated deployment threshold.

Finally, the operational groups concern frequency, source/shift, and geographic structure. "Group-preserving" is not a demographic-fairness guarantee, and the method assumes meaningful groups are available when calibrating the selector. More natural-noise datasets, larger verified sets, and full downstream retraining would provide stronger deployment evidence.

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